

DEALFLOW360

Detailed Product Requirements Document (PRD)

Intelligent, Self-Governing B2B Sales Operations Platform

Hackathon: Odoo Hackathon

Constraint: 24-hour implementation

Purpose: Single source of truth for scope, outcomes, workflow, business rules, technical direction, priorities and acceptance criteria.

Version 1.0 — PRD Baseline

1. Executive Summary

DealFlow360 is an intelligent B2B sales operations platform that manages a complete deal from quotation creation through discount governance, approval, upsell/cross-sell, multi-warehouse fulfillment, hybrid billing, customer negotiation, re-approval, payment and deal-health monitoring.

The central business object is the Deal / Quotation. The product is not a collection of disconnected screens; all screens operate on shared persistent business data and represent different stages or views of the same transaction lifecycle.



The PRD is based on the DealFlow360 problem statement, the judges' Excalidraw workflow, and the stated Must Have / Nice to Have requirements supplied for the hackathon.

2. Problem Definition

2.1 Existing Business Problem

Real B2B sales are more complicated than Create Quote → Invoice → Payment. A single deal may require customer-specific discount limits, manager/finance approvals, margin checks, product recommendations, inventory allocation across warehouses, partial fulfillment, subscriptions, customer negotiation and operational monitoring.

2.2 Problems We Solve

- Unauthorized or excessive discounts causing margin leakage.
- Manual and slow approval routing.
- Poor visibility into stalled or risky deals.
- Inefficient warehouse allocation when inventory is distributed across locations.
- Difficulty combining one-time products with recurring services.
- Disconnected customer negotiation and change-request workflows.
- Missed upsell/cross-sell opportunities.
- Lack of a single quotation-to-closure operational view.

2.3 Product Outcome

The system should evaluate a deal, determine what actions are required, route it to the appropriate users, recommend operational choices, persist every important state change, and expose exceptions to the right role.

3. Product Vision, Goals and Outcomes

3.1 Vision

Build a self-governing B2B deal engine that automatically evaluates, routes, monitors and executes sales operations based on configurable business rules.

3.2 Goals

ID	Goal	Expected Outcome
G1	Quotation management	Create and manage multi-line quotations using persistent data.
G2	Discount governance	Automatically check discounts against customer/category limits.
G3	Approval routing	Automatically generate the required manager/finance approval chain.
G4	Upsell/cross-sell	Recommend relevant products/services during quote creation.
G5	Fulfillment intelligence	Allocate stock across warehouses and handle splits/backorders.
G6	Hybrid billing	Support one-time and recurring items in one deal.
G7	Customer negotiation	Provide a secure portal for review, counter-offers and confirmation.
G8	Deal monitoring	Identify stalled, anomalous and delayed deals.
G9	Reporting	Provide dynamic sales and operational visibility.

3.3 Definition of Success

A judge should be able to complete the core workflow entirely through the UI without manually editing database records.

4. Target Users and Roles

Role	Responsibility	Key Access
Sales Representative	Create/manage deals	Customers, quotations, products, discount tracking, fulfillment progress, negotiations
Sales Manager / Approver	Approve governed discounts	Approval queue, risk analysis, approve/reject/revise, deal health
Finance / Operations	Second-level approvals, fulfillment and billing	Approvals, warehouses, backorders, invoices, subscriptions, payments
Customer	Review and negotiate own deals	Restricted portal, quotation, requests, counter discount, confirmation, invoices
Admin	Configure and monitor platform	Products, customers, warehouses, rules, plans, reports

5. Product Scope and Modules

#	Screen / Module	Purpose
1	Login / Signup	Authentication and role access
2	Sales Dashboard	Sales/deal overview
3	Quotations List	Search, filter and manage quotations
4	Quotation Detail	Central deal workspace
5	Approvals List	Approval queue
6	Approval Detail	Risk analysis and approval decision
7	Fulfillment List	Fulfillment overview
8	Fulfillment Detail	Warehouse allocation, split shipment and backorder
9	Subscriptions List	Recurring contracts
10	Billing Detail	Hybrid billing information
11	Customer Portal	Restricted customer quotation workflow
12	Invoices List	Invoice management

#	Screen / Module	Purpose
13	Invoice Detail	Invoice/payment information
14	Deal Health Dashboard	Stalled deals, anomalies and delivery risks
15	Admin Reporting	Business analytics
16	Product Dashboard	Product management
17	Product Details	Product configuration and stock details
18	Discount Tiers & Approval	Discount governance configuration

6. End-to-End Business Workflow

ADMIN SETUP

Products + Customers + Warehouses + Stock

Discount tiers + Approval rules + Subscription plans

↓

SALES REP

Create quotation

↓

QUOTE ENGINE

Products + quantities + prices + discounts + tax + margin

↓

DISCOUNT / RISK ENGINE

Customer tier + category limits + blended risk

↓

APPROVAL ENGINE

No approval / Manager / Manager + Finance

↓

UPSELL / CROSS-SELL

Recommendation + margin impact

↓

FULFILLMENT ENGINE

Warehouse allocation + split + backorder

↓

BILLING ENGINE

One-time invoice + recurring subscription

↓

CUSTOMER PORTAL

Review + requests + counter discount

↓

RE-EVALUATION

Counter discount checked again; approval can restart

↓

CONFIRMATION → PAYMENT

↓

DEAL HEALTH + REPORTING

The judges' quick test flow follows the same sequence: setup rules/warehouse/subscription, create an over-limit discount, automatic approval, upsell, warehouse split, one-time plus recurring billing, customer negotiation, re-approval, confirmation, payment and updated invoice status.

7. Functional Requirements

FR-01 — Authentication & RBAC

- Authenticated access to protected modules.
- Every user has a role.
- Navigation and backend permissions reflect the role.
- Customer access is isolated to that customer's records.

FR-02 — Customer Management

- Create, edit, archive and view customers.

- Store name, email, phone, address and customer tier.
- Customer tier participates in discount governance.
- Persist all records in the database.

FR-03 — Product Management

- Create, edit, archive and view products.
- Fields: name, category, type, selling price, cost, billing type.
- Types: Goods, Service, Combo.
- Billing: One-time or Recurring.
- Cost is required for margin calculation.

FR-04 — Warehouse & Stock

- Configure warehouses and locations.
- Maintain product stock per warehouse.
- Store data for fulfillment/shipping cost estimation.
- Use persistent stock data for allocation.

FR-05 — Quotations

- Create quotation for a customer.
- Add products and quantities.
- Apply unit prices, discounts and tax.
- Calculate subtotal, discount, tax, total, cost, margin and margin percentage.
- Persist quotation status and history.

FR-06 — Discount Governance

- Configure customer tiers such as Bronze, Silver and Gold.
- Configure category/product discount ceilings.
- Evaluate every quotation line.
- Detect overall and blended violations.
- Calculate a real risk result.
- Create approval requirements from configured rules rather than hardcoded demo values.

FR-07 — Approval Management

- Automatically create approval requests when required.
- Determine manager vs manager+finance chain.
- Support approve, reject and revision request.
- Record approval actions in audit history.
- Prevent unauthorized approval.

FR-08 — Upsell/Cross-sell

- Show recommendations during quote creation.
- Use explainable rule-based recommendations for MVP.
- Show reason, price and margin impact.
- Adding a recommendation immediately updates the quote.

FR-09 — Fulfillment

- Check stock after the appropriate approval stage.
- Recommend warehouse allocation.
- Split orders across warehouses.
- Allow controlled manual override.
- Create backorders when stock is insufficient.
- Track fulfillment status, shipment count and estimated cost.

FR-10 — Subscription & Hybrid Billing

- Allow one-time and recurring products in one quotation.
- Support monthly, quarterly and yearly plans.
- Create subscription records for recurring lines.
- Create invoice records for one-time lines.
- Support simplified day-based proration for MVP.
- Support practical cancellation/modification/refund/credit-note concepts where feasible.

FR-11 — Invoice & Payment

- Create and list invoices.
- Store invoice number, customer, quotation, items, amount, tax and due date.
- Statuses: Draft, Issued, Partially Paid, Paid, Overdue.
- Record Cash or Bank payments internally.
- Update payment and invoice status dynamically.

FR-12 — Customer Portal & Negotiation

- Separate restricted customer-facing view.
- Customer sees only own quotations/invoices.
- View line items and totals.
- Add line-level comments/change requests.
- Submit counter discount.
- Re-enter approval when the counter discount violates rules.
- Confirm quotation after required approvals.

FR-13 — Deal Health

- Detect stalled deals.
- Detect discount anomalies.
- Detect delivery slippage.
- Generate actionable alerts/nudges.
- Allow navigation from alert to relevant deal.

FR-14 — Reporting

- Dynamic sales, approval, fulfillment, billing and health metrics.
- Practical date/status filters where feasible.
- Reports derived from database data, not static JSON.

FR-15 — Audit Trail

- Record major state-changing actions.
- Store actor, action, timestamp, entity and relevant before/after state.
- Cover discount changes, approvals, negotiations, fulfillment, invoice generation and payments.

8. Business Rules and Decision Engines

8.1 Quotation Calculation

Subtotal = $\Sigma(\text{quantity} \times \text{unit price})$
Discount Amount = applicable discount
Taxable Amount = Subtotal – Discount
Total = Taxable Amount + Tax
Cost = $\Sigma(\text{quantity} \times \text{product cost})$
Gross Margin = Total – Cost
Margin % = $\text{Gross Margin} / \text{Total} \times 100$

8.2 Discount Governance

1. Load customer tier.
2. Load applicable tier/category limits.
3. Evaluate every quotation line.
4. Identify violations.
5. Calculate blended risk.
6. Determine approval chain.
7. Persist evaluation and approval request.

8.3 Blended Risk

Use a deterministic, explainable scoring approach for the hackathon. The important requirement is that risk is calculated from actual deal data and rules.

Condition	Illustrative Result
No violation	Low
Small single violation	Medium
Large violation	High
Multiple violations	High / Critical
Negotiated discount creates violation	Re-evaluate and route again

Example: Gold customer may have a 15% overall limit; Hardware may allow 15% while Service allows 10%. A Laptop at 12% is within its limit, while a Service line at 18% violates the Service limit and flags the quote. The blended approach can also aggregate smaller violations across multiple lines.

8.4 Approval Routing

Within permitted range → No approval
 Above first threshold → Manager approval
 Above higher threshold / high risk → Manager → Finance

Thresholds should be configurable in the database.

8.5 Upsell Engine

IF Laptop added → Warranty / Mouse / Keyboard
 IF Server added → Support / Backup Service
 Only recommend active rules/products and optional margin-qualified recommendations.

8.6 Warehouse Allocation

Input: order quantity + stock by warehouse
 Output: allocation(s) + remaining quantity + backorder
 + shipment count + estimated fulfillment cost

Example:
 Order 10; A=6; B=4 → A=6, B=4, Backorder=0
 Order 15; A=6; B=4 → A=6, B=4, Backorder=5

8.7 Billing Engine

ONE_TIME item → invoice
 RECURRING item → subscription
 Mixed quote → one-time invoice + recurring subscription

8.8 Deal Health Rules

Approval pending > threshold → STALLED
 Fulfillment beyond delivery threshold → DELIVERY_RISK
 Discount anomaly → DISCOUNT_ANOMALY
 Customer negotiation delayed → NEGOTIATION_DELAY

9. State Machines

9.1 Quotation Lifecycle

DRAFT
 ↓

```

SUBMITTED
↓
Approval required?
├ No → APPROVED
└ Yes → PENDING_MANAGER
↓
APPROVED / REJECTED / REVISION_REQUESTED
↓
PENDING_FINANCE (if required)
↓
APPROVED
↓
READY_FOR_FULFILLMENT
↓
FULFILLMENT
├ COMPLETE
└ PARTIAL → BACKORDER
↓
BILLING
↓
CUSTOMER_CONFIRMATION
↓
CLOSED

```

9.2 Negotiation Lifecycle

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Quoted
↓
Customer reviews
↓
No change? — Yes → Confirm
|
No
↓
Counter request
↓
Discount re-evaluation
↓
Approval required?
├ No → Customer confirm
└ Yes → Approval chain → Approved → Customer confirm

```

9.3 State Transition Principle

A button must not merely change a label. A state transition should validate the user, execute business rules, persist the change, record an audit event, and return the updated state to the UI.

10. Data Requirements

Group	Core Entities
Identity	User, Role, Permission
Master Data	Customer, Product, Category, Warehouse, Warehouse Stock
Sales	Quotation, Quotation Item
Governance	Discount Tier, Discount Rule, Approval Rule, Approval, Approval Action
Recommendations	Upsell Rule, Cross-sell Rule
Fulfillment	Fulfillment, Fulfillment Item, Backorder
Billing	Subscription Plan, Subscription, Invoice, Invoice Item, Payment
Customer Collaboration	Negotiation, Negotiation Message
Monitoring	Deal Health Event, Notification, Audit Log

10.1 Core Relationships

Customer → Quotation → Quotation Items → Products
Quotation → Discount / Approval
Quotation → Fulfillment → Warehouse / Stock
Quotation → Invoice / Subscription → Payment
Quotation → Negotiation → possible re-approval
Product → Upsell/Cross-sell Rules
Major state changes → Audit Log

10.2 Dynamic Data Principle

The normal data path must be UI → backend/API → database → updated UI. Seed data is acceptable for initial setup and demo readiness, but the platform must create, update and retrieve real persistent records.

11. Technical Approach

11.1 Recommended Architecture

Frontend: React / Next.js + TypeScript + responsive UI
↓
Application / API Layer
↓
PostgreSQL / Supabase
├ Authentication
├ Persistent relational data
└ Realtime where useful

The stack should favor technologies the team already understands. For this 24-hour project, reliability and integration are more important than introducing unfamiliar trendy technologies.

11.2 Core Architecture Principle

Real database + real business logic + real state transitions. The business rules must not be faked in the UI.

11.3 AI Strategy

- Do not force AI into every feature.
- Use deterministic rules for discount governance, approval routing, upsell and deal health in the MVP.
- If the core system is stable, optionally add an AI explanation layer such as 'Why is this deal high risk?' as a secondary feature.
- Any AI-generated code or logic must be understood, tested and adapted by the team.

12. UI/UX Requirements

12.1 Navigation

DEALFLOW360
├ Dashboard
├ Sales
│ ├ Quotations
│ └ Customers
├ Approvals
├ Operations
│ ├ Fulfillment
│ ├ Warehouses
│ └ Backorders
├ Billing
│ ├ Invoices
│ └ Subscriptions
├ Intelligence
│ └ Deal Health
├ Products
├ Administration
├ Discount Rules
├ Approval Rules
└ Reports

12.2 UI Principles

- Responsive layouts.
- Consistent colors, typography, spacing, buttons, cards, tables and status indicators.
- Intuitive menu placement and clear primary actions.
- Useful search/filter/sort on lists where appropriate.
- Loading, empty, success, error and confirmation states.
- Visible approval, fulfillment, payment and risk status.
- Readable validation messages.

12.3 Customer UI Isolation

Customer screens must expose only customer-safe information. Internal margin, approval notes, warehouse configuration and admin reporting must remain unavailable.

13. Security, Validation and Non-Functional Requirements

13.1 Security

- Authentication for protected routes.
- Backend/server-side role enforcement.
- Customer records isolated by authenticated identity.
- Internal margin and approval information hidden from customers.
- Approval actions verify user authority.

13.2 Input Validation

- Quantity > 0.
- Price/cost valid and non-invalid.
- Discount within system-valid bounds.
- Fulfillment allocation ≤ available stock.
- Required fields present.
- Email and structured inputs validated.
- Counter discounts always re-evaluated.

13.3 Reliability

- Avoid partial inconsistent state changes.
- Use appropriate transactional/idempotent operations.
- Record important state changes in audit logs.
- Return user-friendly business errors.

13.4 Performance

- Avoid unnecessary requests.
- Use filtered/indexed database queries.
- Keep dashboards responsive with realistic seed data.

14. Judges' Requirements Alignment

Requirement	Implementation Response
Dynamic data	Database-backed persistent records; static JSON only for initial seed/prototyping.
Responsive/clean UI	Consistent responsive design system.
Robust validation	Frontend + backend/business-rule validation.

Requirement	Implementation Response
Intuitive navigation	Role-aware navigation grouped by business function.
Version control	Git collaboration with meaningful commits/branches and contributions from multiple members.
Backend APIs / data model	Explicit API/service layer and relational model.
Local/offline consideration	Minimize unnecessary external dependencies; keep core rules in the application.
Understand AI/code	Use explainable rules first; AI/code must be understood and adapted.
Trendy technologies	Use modern tools only when they add real value and are reliable within the deadline.

15. MVP and Prioritization

P0 — Critical End-to-End

- Authentication/RBAC
- Products and customers
- Quotation creation/calculation
- Discount governance
- Approval workflow
- Warehouse/stock
- Fulfillment and backorder
- Invoice/payment
- Customer portal and negotiation

P1 — Important

- Upsell/cross-sell
- Subscriptions and hybrid billing
- Deal Health
- Audit trail
- Admin reporting

P2 — Polish

- Advanced charts
- Animations
- Advanced filters/search
- Extra notifications
- Optional AI explanations
- Advanced analytics

15.1 Out of Scope for 24-Hour MVP

- Full ERP
- Complex accounting system
- Real payment gateway
- Real shipping-provider integration
- Advanced ML recommendation model
- Complex tax compliance
- Multi-country taxation
- Enterprise multi-company architecture
- Sophisticated forecasting
- Large-scale edge-case subscription accounting

16. 24-Hour Implementation Approach

Phase 1 — Foundation

- Project setup
- Database schema
- Authentication
- Roles/permissions
- Application shell/navigation

Phase 2 — Core Sales

- Customers
- Products
- Quotation
- Quotation items
- Totals and margin

Phase 3 — Intelligence

- Discount engine
- Risk score
- Approval engine
- Upsell rules

Phase 4 — Operations

- Warehouses
- Stock
- Fulfillment allocation
- Split fulfillment
- Backorders

Phase 5 — Billing

- Subscription plans
- Subscriptions
- Invoices
- Payments
- Hybrid billing

Phase 6 — Customer

- Customer authentication
- Restricted portal
- Negotiation
- Counter discount
- Re-approval

Phase 7 — Monitoring

- Deal Health
- Alerts/notifications
- Admin reporting
- Audit trail

Phase 8 — UI and Reliability

- Responsive polish
- Validation

- Loading/error/empty states
- Seed/demo data
- Navigation consistency

Phase 9 — Final Judge Test

Run the complete golden flow without manually modifying database records. Fix integration failures before spending remaining time on visual polish.

17. Testing and Acceptance Criteria

ID	Acceptance Criterion
AC-01	User logs in and receives correct role.
AC-02	Admin configures products, customers, warehouses and discount rules.
AC-03	Sales rep creates a quotation from real database data.
AC-04	Discounts are evaluated automatically.
AC-05	Approval requirements are generated automatically.
AC-06	Unauthorized users cannot approve restricted quotations.
AC-07	Upsell recommendations can be added.
AC-08	Quote total and margin update immediately.
AC-09	Warehouse stock is checked from persistent inventory.
AC-10	Order can be split across warehouses.
AC-11	Insufficient stock creates a backorder.
AC-12	One-time and recurring items coexist in one quotation.
AC-13	Subscription record is created for recurring items.
AC-14	Customer sees only their own portal data.
AC-15	Customer can submit a counter discount.
AC-16	Negotiated discount is evaluated again.
AC-17	Invoice/payment status updates correctly.
AC-18	Deal Health identifies configured anomalies.
AC-19	Dashboard metrics come from dynamic database data.
AC-20	Complete workflow works without database manipulation.

18. Recommended Judge Demo Scenario

Customer: ABC Corporation

Products:

10 × Laptop

1 × Installation Service

1 × Monthly Support

1. Sales rep creates quotation.
2. Apply an over-limit discount.
3. Discount engine detects violation.
4. Manager approval is created.
5. Manager approves.
6. System recommends Warranty / related product.
7. Sales rep adds it; total and margin update.
8. Fulfillment checks stock:
Warehouse A → 6
Warehouse B → 4

9. Billing:

Laptop + Installation → One-time invoice

Support → Recurring subscription

10. Customer opens restricted portal.

11. Customer submits a higher counter discount.

12. Discount engine re-evaluates.

13. Approval restarts if required.

14. Manager/Finance approves.

15. Customer confirms.

16. Payment is recorded.

17. Deal Health and Reporting update.

This scenario deliberately mirrors the judges' quick test flow so the demonstration proves connected business logic rather than isolated UI screens.

19. Success Outcomes

- A judge can complete the core deal lifecycle entirely through the platform.
- Discount governance and approval routing are visibly automatic.
- Database state changes after important actions.
- The customer portal is genuinely restricted.
- Warehouse split and backorder behavior are demonstrable.
- One-time and recurring billing are visibly separated.
- Negotiation can trigger re-approval.
- Deal Health reflects actual transaction conditions.
- Reports reflect real dynamic data.
- The UI is responsive, consistent and understandable.
- The platform demonstrates real business logic rather than a collection of static screens.

20. Future Roadmap

- Advanced AI sales recommendations.
- Historical purchase-based recommendation models.
- Predictive deal-health and risk scoring.
- Email and notification integrations.
- PDF quotations and richer documents.
- Payment gateway integration.
- Shipping/logistics integrations.
- Multi-company and multi-currency support.
- Advanced taxation and compliance.
- Advanced analytics and forecasting.

21. Final Product Definition

DealFlow360 is a rule-driven B2B sales operations platform that automatically governs discounts and approvals, recommends additional products, optimizes fulfillment across warehouses, handles hybrid billing, enables customer negotiation, and continuously monitors deal health through one connected quotation-to-closure workflow.

21.1 Core Development Principle

No feature is complete until it works from UI → backend → database → business logic → updated UI.

BAD:

Click Approve → UI changes to "Approved"

GOOD:

Click Approve

↓
Backend verifies approver
↓
Approval/state rules validated
↓
Action persisted
↓
Quotation status updated
↓
Next approval step created if required
↓
Audit event recorded
↓
Updated state returned to UI

21.2 Development Pipeline

PRD
↓
System Architecture
↓
Database Design
↓
Business Rules
↓
API / Backend Design
↓
UI/UX Design
↓
Module Development
↓
Testing + Integration
↓
End-to-End Test
↓
Judge Test Flow
↓
Demo Ready

This PRD is the product baseline. The next document should translate it into the Technical Design Document: exact database schema, relationships, permissions, state machines, APIs, algorithms, frontend structure and the detailed 24-hour execution plan.